ADVANCED 5G AND 6G COMMUNICATION SYSTEM

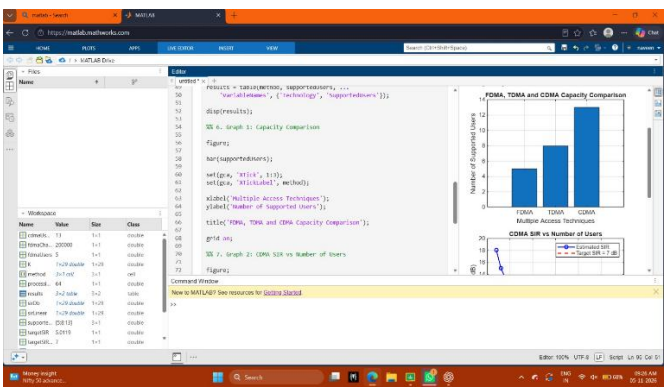
EXPERIMENT 1

PROGRAM CODE:

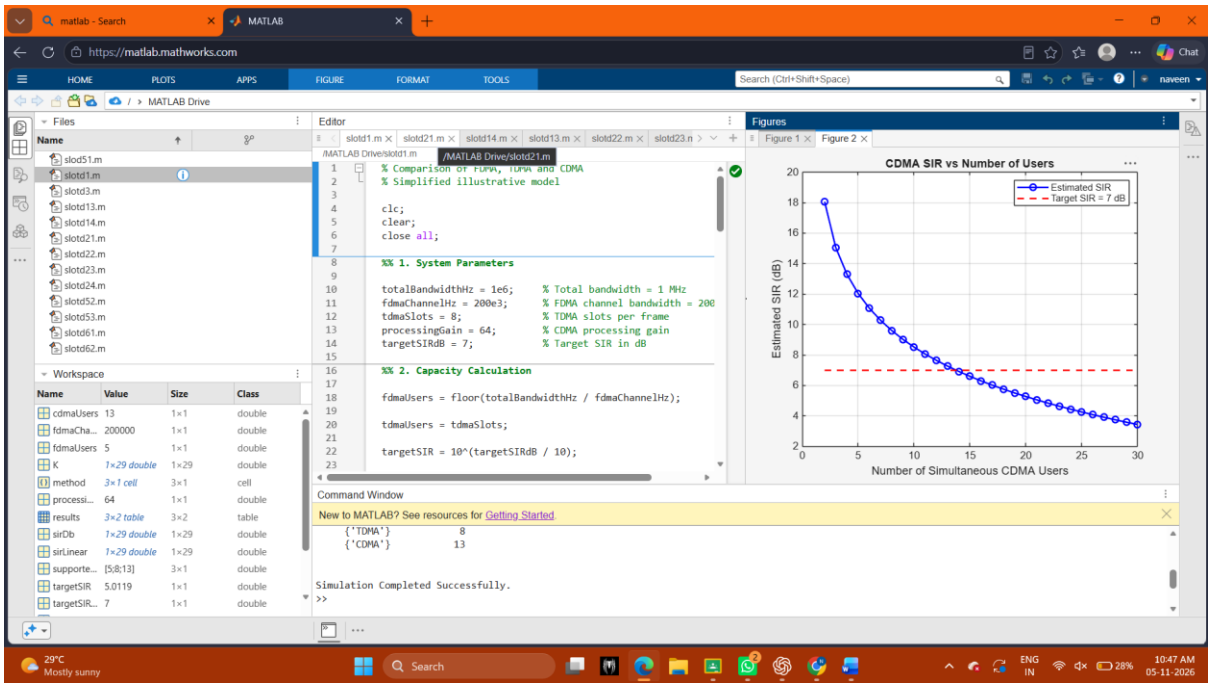
```
clear; clc; close all;
% Illustrative system parameters
totalBandwidthHz = 1e6; % Total bandwidth: 1 MHz
fdmaChannelHz = 200e3; % FDMA channel width: 200 kHz
tdmaSlots = 8; % Number of TDMA slots per frame
processingGain = 64; % CDMA processing gain
targetSIRdB = 7; % Minimum acceptable CDMA SIR
% Capacity estimates under the stated assumptions
fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);
tdmaUsers = tdmaSlots;
targetSIR = 10^(targetSIRdB / 10);
cdmaUsers = floor(1 + processingGain / targetSIR);
fprintf('Illustrative maximum supported users\n');
fprintf('FDMA: %d users\n', fdmaUsers);
fprintf('TDMA: %d users\n', tdmaUsers);
fprintf('CDMA: %d users at a target SIR of %.1f dB\n\n', ...
cdmaUsers, targetSIRdB);
% CDMA SIR as the number of simultaneous users increases
K = 2:30;
sirLinear = processingGain ./ (K - 1);
sirDb = 10 * log10(sirLinear);
% Display a comparison table
method = ["FDMA"; "TDMA"; "CDMA"];
supportedUsers = [fdmaUsers; tdmaUsers; cdmaUsers];
```

```
results = table(method, supportedUsers);
disp(results);
% Plot estimated user capacity
figure;
bar(categorical(method), supportedUsers);
ylabel('Illustrative maximum supported simultaneous users');
title('FDMA, TDMA and CDMA: Capacity Estimate');
grid on;
% Plot estimated CDMA SIR as users are added
figure;
plot(K, sirDb, 'o-', 'LineWidth', 1.5);
hold on;
yline(targetSIRdB, '--r', 'Target SIR');
xlabel('Number of simultaneous CDMA users (K)');
ylabel('Estimated SIR (dB)');
title('Estimated CDMA SIR versus Number of Users');
grid on;
hold off;
```

EXPECTED OUTPUT:



OBJECTIVE 2:



OBJECTIVE 3:

PROGRAM CODE:

```
generation = ["1G"; "2G"; "2G"; "3G"; "4G"; "5G"];

system = ["AMPS"; "GSM"; "IS-95"; "UMTS"; "LTE"; "5G NR"];

accessMethod = ["FDMA"; "TDMA"; "CDMA"; "WCDMA"; "OFDMA"; "OFDMA-based"];

keyIdea = ["Separate frequency channels"; ...

"Separate time slots"; ...

"Code-based user separation"; ...

"Wideband code-based access"; ...

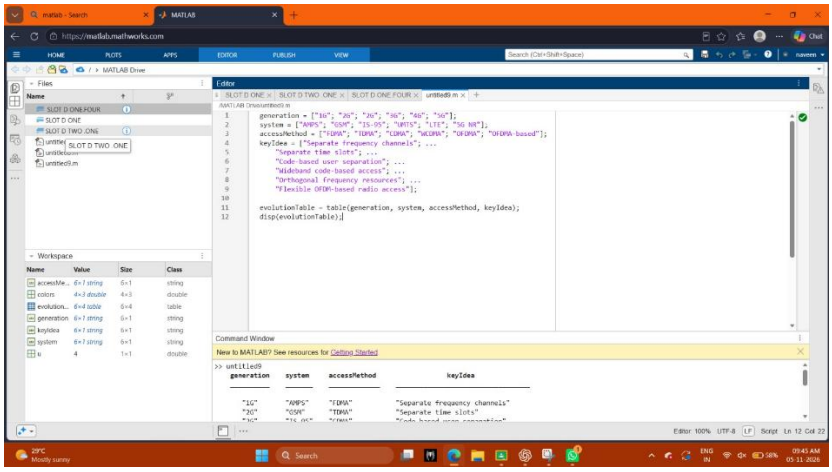
"Orthogonal frequency resources"; ...

"Flexible OFDM-based radio access"];


evolutionTable = table(generation, system, accessMethod, keyIdea);

disp(evolutionTable);
```

EXPECTED OUTPUT:



OBJECTIVE 4:

```
nexttile;
hold on;
colors = lines(4);
for u = 1:4
rectangle('Position',[0, u-1, 1, 0.8], ...
'FaceColor',colors(u,:), 'EdgeColor','k');
text(0.5, u-0.6, sprintf('User %d',u), ...
'HorizontalAlignment','center');
end
xlim([0 1]); ylim([0 4]);
xlabel('Time');
ylabel('Frequency channel');
title('FDMA');

set(gca,'YTick',[]);
hold off;
```

```
% TDMA: users occupy different time slots on one frequency channel
nexttile;
hold on;
for u = 1:4
rectangle('Position',[u-1, 0, 0.8, 1], ...
'FaceColor',colors(u,:), 'EdgeColor','k');
text(u-0.6, 0.5, sprintf('User %d',u), ...
'HorizontalAlignment','center');
end
xlim([0 4]); ylim([0 1]);
xlabel('Time slot');
ylabel('Frequency channel');
title('TDMA');
set(gca,'YTick',[]);
hold off;
% CDMA: users share the same time and frequency band using different codes
nexttile;
hold on;
for u = 1:4
rectangle('Position',[0, 0, 1, 1], ...
'FaceColor',colors(u,:), 'FaceAlpha',0.22, ...
'EdgeColor',colors(u,:), 'LineWidth',1.5);
end
text(0.5, 0.62, 'Users 1-4 share', 'HorizontalAlignment','center');
text(0.5, 0.40, 'band; different codes', 'HorizontalAlignment','center');
xlim([0 1]); ylim([0 1]);
xlabel('Time');
ylabel('Frequency band');
```

```
title('CDMA');
set(gca,'YTick',[]);
hold off;
sgtitle('Conceptual Time-Frequency Resource Allocation'):
```

EXPECTED OUTPUT:

